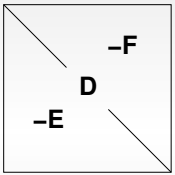
# Iterative methods: relaxation techniques and projection methods

- Basic relaxation methods: Jacobi, Gauss-Seidel, SOR
- Introduction to projection-type techniques
- Sample one-dimensional Projection methods
- Some theory and interpretation –
- See Chapter 4 and Chapter 5 of text for details.

# Linear Systems: Basic Relaxation Schemes

Relaxation schemes: based on the decomposition $A = D - E - F$ where $D$ = diag(A), $-E$ = strict lower part of $A$ and $-F$ its strict upper part.




- For example, Gauss-Seidel iteration :

$$(D - E)x^{(k+1)} = Fx^{(k)} + b$$

- Most common techniques 60 years ago. Now: used as smoothers in Multigrid or as preconditioners

✍ 1 If $\rho_i^{(k)} = i$th component of current residual $b - Ax$ then GS can be written in 'relaxation' form as:

$$\xi_i^{(k+1)} = \xi_i^{(k)} + \frac{\rho_i^{(k)}}{a_{ii}}$$

for $i = 1, \cdots, n$

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# Iteration matrices

✦ Jacobi, Gauss-Seidel, SOR, & SSOR iterations are of the form $\boxed{x^{(k+1)} = Mx^{(k)} + f}$

- $M_{Jac} = D^{-1}(E+F) = I - D^{-1}A$
- $M_{GS}(A) = (D-E)^{-1}F = I - (D-E)^{-1}A$

*SOR* relaxation: $\boxed{\xi_i^{(k+1)} = \omega \xi_i^{(GS,k+1)} + (1-\omega)\xi_i^{(k)}}$

- $M_{SOR}(A) = (D-\omega E)^{-1}(\omega F + (1-\omega)D)$
  $= I - (\omega^{-1}D - E)^{-1}A$

✍ 2 Matlab: take a look at: *gs.m*, *sor.m*, and *sorRelax.m* in iters/

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# An observation & Introduction to Preconditioning

- The iteration $x^{(k+1)} = Mx^{(k)} + f$ is attempting to solve $(I - M)x = f$.
- Since $M = I - P^{-1}A$ this system can be written as: $P^{-1}Ax = P^{-1}b$ ('Preconditioned system')

where for SSOR, we have $P_{SSOR} = (D - \omega E)D^{-1}(D - \omega F)$

- Referred to as the SSOR 'preconditioning' matrix.

In other words: *Relaxation Scheme* $\Longleftrightarrow$ *Preconditioned Fixed Point Iteration*

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# Projection Methods

- The main idea of projection methods is to extract an approximate solution from a subspace.
- We define a subspace of approximants of dimension $m$ and a set of $m$ conditions to extract the solution
- These conditions are typically expressed by orthogonality constraints.
- This defines one basic step which is repeated until convergence (alternatively the dimension of the subspace is increased until convergence).

Example: Each relaxation step in Gauss-Seidel can be viewed as a projection step

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# Background on projectors

- A projector is a linear operator that is idempotent: $\boxed{P^2 = P}$
  Any decomposition $\mathbb{R}^n = K \oplus S$ defines a (unique) projector $P$:
  - From $x = x_1 + x_2$, set $Px = x_1$. In this case:
  - **Ran**$(P) = K$ and **Null**$(P) = S$; $dim(K) = m \rightarrow dim(S) = n - m$.
- Pb: express mapping $x \rightarrow u = Px$ in terms of $K, S$
- Note $u \in K$, $x - u \in S$
- Express 2nd part with $m$ constraints: let $L = S^{\perp}$, then

$$\boxed{u = Px \text{ iff } \begin{cases} u \in K \\ x - u \perp L \end{cases}}$$

✦ Projection onto $K$ and orthogonally to $L$

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# Background on projectors

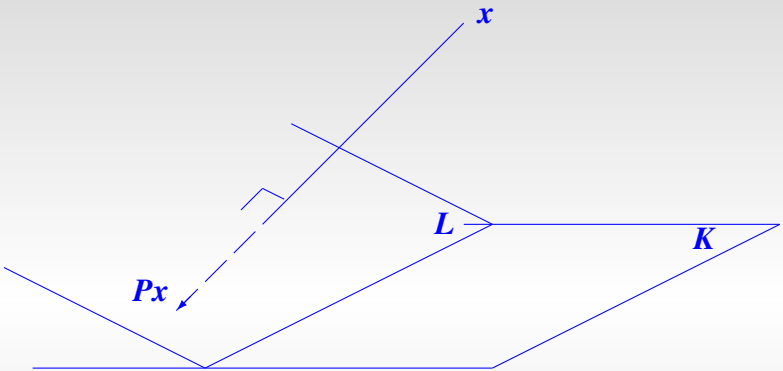
- A projector is a linear operator that is idempotent: $\boxed{P^2 = P}$

*A few properties:*

1. $P$ is a projector iff $I - P$ is a projector
2. $x \in \textbf{Ran}(P)$ iff $x = Px$ iff $x \in \textbf{Null}(I - P)$
3. This means that : $\textbf{Ran}(P) = \textbf{Null}(I - P)$ .
4. Any $x \in \mathbb{R}^n$ can be written (uniquely) as $x = x_1 + x_2$,
   $x_1 = Px \in \textbf{Ran}(P)$ $x_2 = (I - P)x \in \textbf{Null}(P)$
   So: $\boxed{\mathbb{R}^n = \textbf{Ran}(P) \oplus \textbf{Null}(P)}$

✍ 3 Prove the above properties

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- Illustration: $P$ projects onto $K$ and orthogonally to $L$
- When $L = K$ projector is orthogonal.
- Note: $Px = 0$ iff $x \perp L$.

# Projection methods for linear systems

- Initial Problem: $b - Ax = 0$
- Given two subspaces $K$ and $L$ of $\mathbb{R}^N$ of dimension $m$, define ...

  Approximate problem: Find $\tilde{x} \in K$ such that $\underbrace{b - A\tilde{x} \perp L}_{\text{Petrov-Galerkin cond.}}$
- $m$ degrees of freedom ($K$) + $m$ constraints ($L$) $\rightarrow$
- To solve: A small linear system ('projected problem')
- Basic projection step. Typically a sequence of such steps are applied

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- With a nonzero initial guess $x_0$, approximate problem is

$$\text{Find} \quad \tilde{x} \in x_0 + K \quad \text{such that} \quad b - A\tilde{x} \perp L$$

Write $\tilde{x} = x_0 + \delta$ and $r_0 = b - Ax_0$. $\rightarrow$ system for $\delta$:

$$\boxed{\text{Find } \delta \ \in K \text{ such that } r_0 - A\delta \perp L}$$

✍ 4 Formulate Gauss-Seidel as a projection method -

✍ 5 Generalize Gauss-Seidel by defining subspaces consisting of 'blocks' of coordinates $\textbf{span}\{e_i, e_{i+1}, \ldots, e_{i+p}\}$

# Matrix representation:

**Setting:**

- $V = [v_1, \ldots, v_m]$ a basis of $K$ &
- $W = [w_1, \ldots, w_m]$ a basis of $L$

- Write approximate solution as $\tilde{x} = x_0 + \delta \equiv x_0 + Vy$ where $y \in \mathbb{R}^m$.

- Then Petrov-Galerkin condition yields: $W^T(r_0 - AVy) = 0$
- Therefore, $\boxed{\tilde{x} = x_0 + V[W^TAV]^{-1}W^Tr_0}$

Remark: In practice $W^TAV$ is known from algorithm and has a simple structure [tridiagonal, Hessenberg,..]

# Prototype Projection Method

## Until Convergence Do:

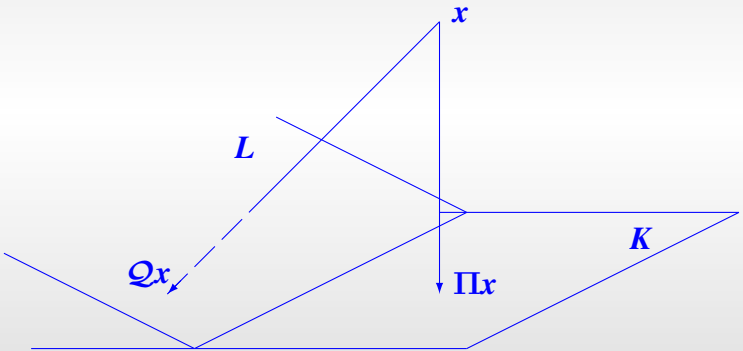
1. Select a pair of subspaces $K$, and $L$;
2. Choose bases: $V = [v_1, \ldots, v_m]$ for $K$ and $W = [w_1, \ldots, w_m]$ for $L$.
3. Compute :

$$r \leftarrow b - Ax,$$
$$y \leftarrow (W^T AV)^{-1} W^T r,$$
$$x \leftarrow x + Vy.$$

# Projection methods: Operator form representation

✦ Let $\Pi$ = orthogonal projector onto $K$, $\mathcal{Q}$ = (oblique) projector onto $K$ and orthogonally to $L$. Assumption: no vector of $K$ is $\perp$ to $L$

$$\Pi x \in K, \quad x - \Pi x \perp K$$
$$\mathcal{Q}x \in K, \quad x - \mathcal{Q}x \perp L$$




$\Pi$ and $\mathcal{Q}$ projectors

In the case $x_0 = 0$, approximate problem amounts to solving

$\mathcal{Q}(b - Ax) = 0, \quad x \in K$ or in operator form (solution is $\Pi x$)

$\mathcal{Q}(b - A\Pi x) = 0$

**Question:** what accuracy can one expect? Let $x^*$ be the exact solution

1) We can't do better than $\|(I - \Pi)x^*\|_2$: $\|\tilde{x} - x^*\|_2 \geq \|(I - \Pi)x^*\|_2$

2) The residual of the exact solution for the approximate problem satisfies:

$\|b - \mathcal{Q}A\Pi x^*\|_2 \leq \|\mathcal{Q}A(I - \Pi)\|_2 \|(I - \Pi)x^*\|_2$

# Two Important Particular Cases.

1. $L = K$

- When $A$ is SPD then $\|x^* - \tilde{x}\|_A = \min_{z \in K} \|x^* - z\|_A$.
- Class of Galerkin or Orthogonal projection methods
- Important member of this class: Conjugate Gradient (CG) method

2. $L = AK$ .

In this case $\|b - A\tilde{x}\|_2 = \min_{z \in K} \|b - Az\|_2$

- Class of Minimal Residual Methods: CR, GCR, ORTHOMIN, GMRES, CGNR, ...

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# One-dimensional projection processes

$$K = span\{d\}$$
and
$$L = span\{e\}$$

Then $\tilde{x} = x + \alpha d$. Condition $r - A\delta \perp e$ yields $\alpha = \frac{(r,e)}{(Ad,e)}$

- Three popular choices:
  - (1) Steepest descent
  - (2) Minimal residual iteration
  - (3) Residual norm steepest descent

*1. Steepest descent.* A is SPD. Take at each step $d = r$ and $e = r$.

Iteration:

$$
\begin{aligned}
&r \leftarrow b - Ax, \\
&\alpha \leftarrow (r, r)/(Ar, r) \\
&x \leftarrow x + \alpha r
\end{aligned}
$$

- Each step minimizes $f(x) = \|x - x^*\|_A^2 = (A(x - x^*), (x - x^*))$ in direction $-\nabla f$.
- Convergence guaranteed if $A$ is SPD.

✍ 6 As is formulated, the above algorithm requires 2 'matvecs' per step. Reformulate it so only one is needed.

*Convergence* based on the Kantorovitch inequality: Let $B$ be an SPD matrix, $\lambda_{max}$, $\lambda_{min}$ its largest and smallest eigenvalues. Then,

$$\frac{(Bx,x)(B^{-1}x,x)}{(x,x)^2} \leq \frac{(\lambda_{max}+\lambda_{min})^2}{4\,\lambda_{max}\lambda_{min}}, \quad \forall x \neq 0.$$

- This helps establish the convergence result

Let $A$ an SPD matrix. Then, the $A$-norms of the error vectors $d_k = x_* - x_k$ generated by steepest descent satisfy:

$$\|d_{k+1}\|_A \leq \frac{\lambda_{max}-\lambda_{min}}{\lambda_{max}+\lambda_{min}}\|d_k\|_A$$

- Algorithm converges for any initial guess $x_0$.

*Proof:* Observe $\|d_{k+1}\|_A^2 = (Ad_{k+1}, d_{k+1}) = (r_{k+1}, d_{k+1})$

- by substitution,

$$\|d_{k+1}\|_A^2 = (r_{k+1}, d_k - \alpha_k r_k)$$

- By construction $r_{k+1} \perp r_k$ so we get $\|d_{k+1}\|_A^2 = (r_{k+1}, d_k)$. Now:

$$\begin{aligned} \|d_{k+1}\|_A^2 &= (r_k - \alpha_k A r_k, d_k) \\ &= (r_k, A^{-1} r_k) - \alpha_k (r_k, r_k) \\ &= \|d_k\|_A^2 \left(1 - \frac{(r_k, r_k)}{(r_k, Ar_k)} \times \frac{(r_k, r_k)}{(r_k, A^{-1} r_k)}\right). \end{aligned}$$

Result follows by applying the Kantorovich inequality. ■

## 2. Minimal residual iteration.

A positive definite ($A + A^T$ is SPD). Take at each step $d = r$ and $e = Ar$.

Iteration:

$$
\begin{aligned}
&r \leftarrow b - Ax, \\
&\alpha \leftarrow (Ar, r)/(Ar, Ar) \\
&x \leftarrow x + \alpha r
\end{aligned}
$$

- Each step minimizes $f(x) = \|b - Ax\|_2^2$ in direction $r$.
- Converges under the condition that $A + A^T$ is SPD.

✍ 7 As is formulated, the above algorithm would require 2 'matvecs' at each step. Reformulate it so that only one matvec is required

## Convergence

Let $A$ be a real positive definite matrix, and let

$$\mu = \lambda_{min}(A + A^T)/2, \quad \sigma = \|A\|_2.$$

Then the residual vectors generated by the Min. Res. Algorithm satisfy:

$$\|r_{k+1}\|_2 \leq \left(1 - \frac{\mu^2}{\sigma^2}\right)^{1/2} \|r_k\|_2$$

- In this case Min. Res. converges for any initial guess $x_0$.

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*Proof:* Similar to steepest descent. Start with

$$\begin{aligned} \|r_{k+1}\|_2^2 &= (r_{k+1}, r_k - \alpha_k A r_k) \\ &= (r_{k+1}, r_k) - \alpha_k (r_{k+1}, A r_k). \end{aligned}$$

By construction, $r_{k+1} = r_k - \alpha_k A r_k$ is $\perp A r_k$, so:
$\|r_{k+1}\|_2^2 = (r_{k+1}, r_k) = (r_k - \alpha_k A r_k, r_k)$. Then:

$$\begin{aligned} \|r_{k+1}\|_2^2 &= (r_k, r_k) - \alpha_k (A r_k, r_k) \\ &= \|r_k\|_2^2 \left( 1 - \frac{(A r_k, r_k)}{(r_k, r_k)} \frac{(A r_k, r_k)}{(A r_k, A r_k)} \right) \\ &= \|r_k\|_2^2 \left( 1 - \frac{(A r_k, r_k)^2}{(r_k, r_k)^2} \frac{\|r_k\|_2^2}{\|A r_k\|_2^2} \right). \end{aligned}$$

Result follows from the inequalities $(Ax, x)/(x, x) \geq \mu > 0$ and
$\|Ar_k\|_2 \leq \|A\|_2 \, \|r_k\|_2$. ■

## 3. Residual norm steepest descent.

A is arbitrary (nonsingular). Take at each step $d = A^T r$ and $e = Ad$.

Iteration:
$$r \leftarrow b - Ax, d = A^T r$$
$$\alpha \leftarrow \|d\|_2^2 / \|Ad\|_2^2$$
$$x \leftarrow x + \alpha d$$

- Each step minimizes $f(x) = \|b - Ax\|_2^2$ in direction $-\nabla f$.
- Important Note: equivalent to usual steepest descent applied to normal equations $A^T Ax = A^T b$ .
- Converges under the condition that $A$ is nonsingular.

✍ 8 Take a look at *demo1DProj.m* in /iters.